COMPUTER NETWORKS

Assignment Question Paper

Maximum Marks: 50	**Duration: 2 Hours**

General Instructions

1. Attempt all questions.
2. Marks are indicated against each question.
3. Show all working for numerical problems.

Q1. **[6 Marks]**

A LAN has a rated bandwidth of 10 Mbps. During a measurement interval, the network was observed to carry 12,000 frames per minute, where each frame is 10,000 bits long.

(a) Calculate the total number of bits transmitted per minute. **[2]**

(b) Compute the effective throughput of the LAN in bps and in Mbps. **[2]**

(c) Explain why the measured throughput is lower than the rated bandwidth of 10 Mbps. **[2]**

Q2. **[7 Marks]**

The following UDP header was captured in hexadecimal:

`CB84 000D 001C 001C`

(a) Separate the header into its hexadecimal byte pairs and identify the four UDP header fields. **[2]**

(b) Determine the source port, destination port, UDP length and checksum values (give decimal values where applicable). **[3]**

(c) Given that the UDP header is 8 bytes long, calculate the size of the UDP data. **[1]**

(d) State the byte order used for port fields and explain what the Length field represents. **[1]**

Q3. **[5 Marks]**

An organisation is allocated the IP block 17.63.110.114/24.

(a) Write the subnet mask and the network address of the block. **[1]**

(b) Determine the first usable address, the last usable address and the broadcast address. **[3]**

(c) State the valid host range and explain the role of the first and last addresses of the block. **[1]**

Q4. **[6 Marks]**

Draw the layered structure of the TCP/IP model (top to bottom) and briefly explain the function of each layer: Application Layer, Transport Layer, Network Layer and Data Link Layer.

Q5. **[6 Marks]**

(a) Describe the Stop-and-Wait protocol and explain why it is inefficient when the round-trip time (RTT) is large compared with the frame transmission time. **[2]**

(b) Write the formula for link utilization (U) in Stop-and-Wait. **[1]**

(c) Explain how Selective Repeat ARQ overcomes these inefficiencies. **[3]**

Q6. **[4 Marks]**

Explain the working of the Cyclic Redundancy Check (CRC) error-detection technique. Describe how the sender computes the check value, how the receiver detects an error, and state the practical result of using CRC.

Q7. **[5 Marks]**

Compare IPv4 and IPv6 on the following bases:

(a) Address size **[1]**

(b) Header and options **[2]**

(c) Built-in features **[2]**

Q8. **[5 Marks]**

Explain how the Dynamic Host Configuration Protocol (DHCP) works. Your answer should cover:

(a) The role of the DHCP server and its address pool **[1]**

(b) The DORA process **[2]**

(c) The use of an ARP probe before address assignment **[1]**

(d) The role of lease expiration **[1]**

Q9. **[6 Marks]**

(a) Compare TCP and UDP in terms of connection type, reliability, ordering, latency and overhead. **[3]**

(b) For each of the following applications, recommend a suitable transport protocol and justify your choice: **[3]**

(i) Real-time streaming, online gaming and VoIP

(ii) On-demand video (e.g., Netflix, YouTube)

— End of Question Paper —